

```
1 import java.util.*;
2
3 public class Main {
4
5     // Partition function for deterministic Quick Sort
6     static int partition(int[] arr, int p, int r) {
7         int x = arr[r];
8         int i = p - 1;
9
10        for (int j = p; j < r; j++) {
11            if (arr[j] <= x) {
12                i++;
13                int temp = arr[i];
14                arr[i] = arr[j];
15                arr[j] = temp;
16            }
17        }
18
19        int temp = arr[i + 1];
20        arr[i + 1] = arr[r];
21        arr[r] = temp;
22
23        return i + 1;
24    }
25
26    // Deterministic Quick Sort
27    static void quickSort(int[] arr, int p, int r) {
28        if (p < r) {
29            int q = partition(arr, p, r);
30
31            quickSort(arr, p, q - 1);
32            quickSort(arr, q + 1, r);
33        }
34    }
35}
```

```

36 // Randomized Partition
37 static int randomPartition(int[] arr, int p, int r, Random random) {
38
39     // Select a random pivot
40     int randomIndex = p + random.nextInt(r - p + 1);
41
42     // Move random pivot to the last position
43     int temp = arr[randomIndex];
44     arr[randomIndex] = arr[r];
45     arr[r] = temp;
46
47     return partition(arr, p, r);
48 }
49
50 // Randomized Quick Sort
51 static void randomizedQuickSort(int[] arr, int p, int r, Random random) {
52     if (p < r) {
53         int q = randomPartition(arr, p, r, random);
54
55         randomizedQuickSort(arr, p, q - 1, random);
56         randomizedQuickSort(arr, q + 1, r, random);
57     }
58 }
59
60 // Display array
61 static void display(int[] arr) {
62     for (int x : arr) {
63         System.out.print(x + " ");
64     }
65     System.out.println();
66 }
67

```

```
68 public static void main(String[] args) {
69
70     Scanner sc = new Scanner(System.in);
71
72     System.out.print("Enter number of elements: ");
73     int n = sc.nextInt();
74
75     int[] arr1 = new int[n];
76     int[] arr2 = new int[n];
77
78     System.out.println("Enter " + n + " elements:");
79
80     for (int i = 0; i < n; i++) {
81         arr1[i] = sc.nextInt();
82         arr2[i] = arr1[i];
83     }
84
85     // Deterministic Quick Sort
86     quickSort(arr1, 0, n - 1);
87
88     // Randomized Quick Sort
89     Random random = new Random();
90     randomizedQuickSort(arr2, 0, n - 1, random);
91
92     System.out.println("\nSorted array using Deterministic Quick Sort:");
93     display(arr1);
94
95     System.out.println("Sorted array using Randomized Quick Sort:");
96     display(arr2);
97
98     sc.close();
99 }
100 }
```

Enter number of elements: 8

Enter 8 elements:

23 34 54 123 34 56 67676 112

Sorted array using Deterministic Quick Sort:

23 34 34 54 56 112 123 67676

Sorted array using Randomized Quick Sort:

23 34 34 54 56 112 123 67676

...Program finished with exit code 0

Press ENTER to exit console.